

FULL-FLEDGED MERN E-COMMERCE PROJECT

Step-by-Step Team Guide (3 Members) with GitHub Workflow

1. PROJECT OVERVIEW

This document explains how to START, BUILD, COLLABORATE, and DEPLOY a real MERN-stack e-commerce website with payments, authentication, and admin panel. The guide is designed for a team of 3 engineering students.

2. TECH STACK

- Frontend: React (Vite)
- Backend: Node.js + Express
- Database: MongoDB Atlas
- Authentication: JWT + bcrypt
- Payments: Razorpay / Stripe
- Version Control: Git & GitHub
- Deployment: Vercel (Frontend), Render/Railway (Backend)

3. TEAM ROLE DISTRIBUTION (IMPORTANT)

Member 1: Backend Lead (APIs, DB, Auth, Payments)

Member 2: Frontend Lead (UI, Pages, Routing, Cart)

Member 3: Integration & Admin (Admin panel, testing, deployment)

4. DAY 1–2: PROJECT INITIALIZATION

Step 1: Create GitHub Repository

- One member creates a GitHub repo (e.g., mern-ecommerce)
- Add README.md and .gitignore (Node)
- Add all 3 members as collaborators

Step 2: Local Setup

- Install Node.js, Git, VS Code
- Create main folder structure:
ecommerce/ → client/ (React) + server/ (Node)

5. DAY 3–5: BACKEND FOUNDATION

- Initialize server with npm init
- Install express, mongoose, dotenv, cors
- Connect MongoDB Atlas
- Create models: User, Product, Order
- Push backend base code to GitHub

6. DAY 6–8: AUTHENTICATION & PRODUCTS

- User register & login APIs
- JWT authentication middleware
- Product CRUD APIs
- Admin-only routes

7. DAY 9–11: FRONTEND DEVELOPMENT

- Setup React routing
- Pages: Home, Shop, About, Contact
- Policy pages: Privacy, Refund, Shipping, Terms
- Product listing & detail pages
- Push frontend code regularly

8. DAY 12–14: CART, ORDERS & PAYMENTS

- Cart using Context API
- Checkout page
- Order creation
- Razorpay/Stripe integration
- Payment success & failure handling

9. DAY 15–16: ADMIN & STATIC CONTENT

- Admin dashboard
- Add/edit products
- Update policy content from DB
- Contact form backend

10. DAY 17–18: SECURITY & TESTING

- Input validation
- Error handling
- JWT protection
- Manual testing (buy product end-to-end)

11. DAY 19–20: DEPLOYMENT

- Deploy backend on Render/Railway
- Deploy frontend on Vercel
- Set environment variables
- Final live testing

12. GITHUB COLLABORATION RULES

- Each feature = new branch
- Use meaningful commit messages
- Daily pull & push
- One member reviews before merge

13. FINAL OUTCOME

At the end of 20 days, your team will have a REAL, production-ready MERN e-commerce website with payments, admin panel, and live deployment. This project is resume-worthy and startup-ready.